Application of FY-4B Geostationary Meteorological Satellite in Grassland Fire Dynamic Monitoring

Jie Chen, Ying Wu, Shuang Wu, Lianni Xie, Jun Tang, Zuomin Xu, Xiuzhen Han, Xiaomin Ma, Wei Zheng, Tao Sun, and Cheng Liu

***Abstract*— In this study, the channel data related to fire point identification of Advanced Geostationary Radiation Imager on Fengyun-4B (FY-4B/AGRI) and Advanced Meteorological Imager on GEO-KOMPSAT 2A (GK-2A/AMI) were cross-compared. A total of 267 sampling points in China (Guangdong, Guangxi, Guizhou, Yunnan, and Hainan) were selected to carry out fine positioning correction on the data in different elevation intervals. Then, a fire monitoring algorithm based on Fengyun-4B (FY-4B) was proposed, in which the self-adaptive threshold adjustment of the underlying surface parameters and the reprocessing module of fire point identification were added. The algorithm can realize the high-precision and stable monitoring of fire. The continuous dynamic monitoring was carried out using the grassland fire Mongolia from April 18 to 20, 2022 as an example. The results showed that the parameters of FY-4B/AGRI and GK2A/AMI channels have high consistency. The root-mean-square error (RMSE) of reflection channel was 1.33%, and the maximum RMSE of brightness temperature channel was less than 1.3 Kelvin (K). Through the positioning analysis in different elevation intervals, the average longitude offset of FY-4B satellite data was −0.5° to 0° and the average latitude offset was −0.9° o 0.6°. Overall, these findings indicate that the high-frequency observations of FY-4B can be fully utilized to monitor forest and grassland fires, which can continuously track the dynamic evolution of fire and can distinguish the spatial distribution of different fire intensities in large-scale fire fields.**

***Index Terms*— Dynamic monitoring, Fengyun-4B (FY-4B) satellite, geostationary meteorological satellite, grassland fire.**

I. Introduction

THE grassland is an important part of terrestrial ecosystem and is currently the most widely distributed vegetation type in the world. It has the functions of water and soil conservation, climate regulation, and biodiversity maintenance [1], [2], [3]. Fire is one of the most important interference factors in terrestrial ecosystem [4], [5], [6]. The grassland fire is a natural disaster that is sudden, destructive, and difficult to manage and rescue [7]. On the one hand, it can cause serious damage to the grassland ecosystem and huge loses of grassland resources. On the other hand, it greatly affects the succession and renewal of vegetation in the natural ecosystem [8].

Satellite remote sensing observation has wide range, high timeliness, and rich information. Due to these natural advantages, the satellite remote sensing observation has been widely used in forest and grassland fire monitoring [9]. The case and algorithm research results of domestic and foreign scholars using Moderate Resolution Imaging Spectroradiometer (MODIS), National Oceanic and Atmospheric Administration (NOAA) series, and other satellites to monitor forest and grassland fire have been widely approved and applied [10], [11], [12], [13]. In recent years, with the launch of new generation geostationary meteorological satellites, such as "Himawari-8" [14], [15] by Japan, Geostationary Operational Environmental Satellites-R/S/T (GOES-R/S/T) by the United States, Fengyun-4A (FY-4A) by China, and GEO-KOMPSAT-2A (GK-2A) by Korea [16], [17], the geostationary satellite data have played a very important role in dynamic monitoring of forest and grassland fire. Wickramasinghe et al. [18] developed a grassland fire monitoring algorithm based on Himawari-8 satellite data for Western Australia and compared it with MODIS fire monitoring data for the same region. The results showed that the daily Advanced Himawari Imager-Fire Surveillance Algorithm (AHI-FSA) omission rate was only 7%. Sometimes, due to the high temporal frequency, AHI-FSA can observe fires in advance of MODIS by 3 h and greatly enhance the details of fire motion. Chen et al. [19] developed an algorithm that is applicable to China based on GK-2A satellite data. The results showed that due to the high timeliness of remote sensing technology, the GK-2A satellite can perform high-frequency and timely observation during fire detection. The multitime and continuous observation can be used to evaluate the influence range and intensity of fires in real time, and we can thus obtain continuous and dynamic fire information, which is of great significance for the real-time observation and rapid assessment of forest fires.

With the development and application of China's new generation of meteorological satellites, the satellite fire monitoring capability of China has been further improved.

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Jie Chen, Xiuzhen Han, Wei Zheng, and Cheng Liu are with the Key Laboratory of Radiometric Calibration and Validation for Environmental Satellites, National Satellite Meteorological Center (National Center for Space Weather), China Meteorological Administration, Beijing 100081, China, and also with the Innovation Center for FengYun Meteorological Satellite (FYSIC), Beijing 100081, China (e-mail: chenjie@cma.gov.cn; zhengw@cma.gov.cn; liucheng@cma.gov.cn).

Ying Wu is with the China Meteorological Administration Public Meteorological Service Center, Beijing 100081, China (e-mail: wuying@cma.cn).

Shuang Wu, Lianni Xie, Zuomin Xu, and Tao Sun are with the Heilongjiang Eco-Meteorology Center, Harbin 150030 China (e-mail: wushuanghlj@163.com; xielianni@163.com; xuzuomin16@mails.ucas.ac.cn; suntao_7675@163.com).

Jun Tang is with State Grid Sichuan Electric Power Company, Gaoxin, Chengdu 610041, China (e-mail: 3201368034@qq.com).

Xiaomin Ma is with the State Grid Sichuan Electric Power Research Institute, Gaoxin, Chengdu 610041, China (e-mail: 373003345@qq.com).

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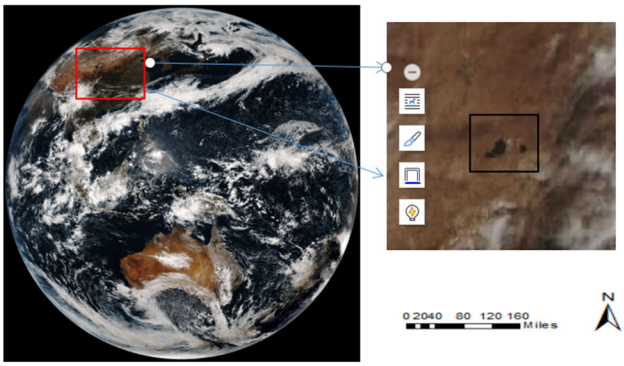


Fig. 1. (a) FY-4B full-disk data. (b) Area of fire.

Chen et al. [20], [21] developed a business algorithm of fire monitoring based on Fengyun-3 (FY-3) series satellites, which has greatly improved the performance and accuracy of the fire monitoring algorithm. However, the low observation frequency of polar-orbit satellites makes it difficult to realize continuous and dynamic monitoring of forest and grassland fire.

Fengyun-4B (FY-4B) is China's latest generation geostationary quantitative remote sensing operational meteorological satellite [22]. It is equipped with the Advanced Geostationary Radiation Imager (AGRI), and the channels have been increased from 5 in FY-2 to 15 in FY-4B. The bands of AGRI cover visible light (VIS), short-wave infrared (SWIR), medium-wave infrared (MWIR), and long-wave infrared (LWIR). The resolution of VIS channel reaches 0.5 km [23]. Compared with FY-4A, FY-4B has achieved great improvement in instruments, calibration, and positioning. It adds a second-level fast scanning mode [24]. The high-quality channel information and high-frequency observation of FY-4B can be widely used in the dynamic monitoring of forest and grassland fire.

II. Data and Methods

A. Study Area

The study area is the FY-4B full-disk data. The Inner Mongolia Autonomous Region of China borders with Mongolia, and the contiguous area has vast grassland where the grassland fire often occurs [7], [25], [26]. In this study, the fire that occurred in the Sino–Mongolian border area from April 18 to 20, 2022 is selected as an example to study the application of FY-4B data in the grassland fire dynamic monitoring (Fig. 1).

B. Instrument Characteristics

The FY-4B was successfully launched at 0:17 Beijing time (BJT) on June 3, 2021 [27]. It is the first operational satellite among China's new-generation geostationary meteorological satellites series. Its subsatellite point is located at 133°E. The FY-4B and FY-4A satellites constitute the China's new-generation geostationary meteorological satellite observation system, realizing a dual satellite network. They jointly monitor the atmosphere and land surface at high frequency and generate various atmospheric physical parameters and quantitative products (cloud top temperature, dust storm detection product, fire hot spot detection, and so on). The parameters of FY-4A and FY-4B are listed in Table I.

TABLE I
Parameters of FY-4A and FY-4B

Satellite	FY-4A FY-4B
Weight	306kg
Number of channels	FY-4A: 14, FY-4B: 15
Spatial resolution	0.5~1 km (VIS/SWIR), 2~4 km (IR)
Time for full-disc scanning	15 min
Time for sector scanning	1min (1000km × 1000km)
Sensitivity	S/N>3 @ 1% (reflection bands) NEΔT: 0.2K @ 300K (IR bands)

VIS:Visible light;SWIR:Short Wave Infrared

The AGRI onboard FY-4B satellite is one of the most important sensors. It has a full aperture and full optical path infrared band intersatellite calibration function and VIS band intersatellite-sunlight calibration function. The radiometric calibration accuracy can reach 3%–5% in the visible band and 0.5 K in the infrared band. Compared with FY-4A, FY-4B adds the second-level fast scanning observation, which can continuously observe for fixed areas. It has one more water vapor band than the FY-4A, reaching 15 observation bands, including three solar reflection bands and 12 infrared bands. Table II lists the parameters and applications of FY-4B/AGRI instrument. In addition, compared with the 5- and 15-min alternating observation modes of FY-4A, FY-4B cancels the 5-min mode and adopts a single 15-min full-disk scanning mode to ensure the uniformity of observation in the time period.

C. Methods

1) FY-4B Data Preprocessing:

a) Reflectivity correction: The solar zenith angle varies within 0°–90° during the FY-4B Earth observation. To obtain more accurate reflectivity, the reflectivity information acquired at different times needs to be corrected. The traditional solar zenith angle correction method is given as follows:

$$R' = R/\cos(\theta_{sz}) \tag{1}$$

where R denotes the original reflectivity of the satellite observation, R' is the reflectivity corrected by the solar zenith angle, and θ_{sz} is the solar zenith angle. When θ_{sz} is small, the reflectivity correction is not affected by the solar zenith angle. When θ_{sz} is close to 90°, the value is infinitely close to 0, resulting in R' anomalies. Therefore, an improved version of the original correction method is needed for large solar zenith angle observations, as in the following equation [19]:

$$R' = R/\cos(\theta_{sz} \times (1.0 - 1.3 \times \sin(0.05 \times \theta_{sz}))). \tag{2}$$

Fig. 2 shows the reflectivity correction results of different correction methods under the observation condition of simulated all solar zenith angles. As can be seen, the correction results of (1) and (2) basically coincide when $\theta_{sz} < 30°$. With the increase of θ_{sz}, the difference in correction results gradually appears, especially after $\theta_{sz} > 60°$. A similar situation can be seen from the ratio (red line) curve of the two methods, that is, the improved reflectivity correction method is more suitable for geostationary satellites.

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TABLE II
PARAMETERS AND APPLICATIONS OF FY-4B/AGRI INSTRUMENT

channel	Central wavelength(μm)	Band(μm)	Spatial resolution(Km)	SNR 0r NEΔT @specified input	
1	0.47	0.45-0.49	1	S/N≥90 (ρ=100 %)	
2	0.65	0.55-0.75	0.5	S/N≥150 (ρ=100 %) @0.5km	S/N≥3 (ρ=1 %) @1km
3	0.825	0.75-0.90	1	S/N≥200 (ρ=100 %)	S/N≥3 (ρ=1 %)
4	1.379	1.371-1.386	2	S/N≥120 (ρ=100 %)	S/N≥2 (ρ=1 %)
5	1.61	1.58-1.64	2	S/N≥200 (ρ=100 %)	S/N≥3 (ρ=1 %)
6	2．225	2.10-2.35	2	S/N≥200 (ρ=100 %)	S/N≥2 (ρ=1 %)
7	3.375	3.50-4.00(high)	2	≤0.7K (315K)	
8	3.375	3.50-4.00(low)	4	0.2K (300K)	2K (240 K)
9	6.25	5.80-6.70	4	0.2K (300K)	0.9K (240 K)
10	6.95	6.75-7.15	4	0.25K(待定) (300K)	0.9K (240 K)
11	7.42	7.24-7.60	4	0.25K (300K)	0.9K (240 K)
12	8.55	8.3-8.8	4	0.2K (300K)	0.4K (240 K)
13	10.80	10.30-11.30	4	0.2K (300K)	0.4K (240 K)
14	12.00	11.50-12.50	4	0.2K (300K)	0.4K (240 K)
15	13.3	13.00-13.60	4	0.5K (300K)	0.9K (240 K)

SNR :Signal-to-noise ratio; NEΔT: Noise equivalent temperature difference in infrared band

b) Projection conversion: The FY-4B geostationary meteorological satellite adopts the geostationary orbit nominal projection defined by the Coordination Group of Meteorological Satellites Low Rate and High Rate Information Transmission (CGMSLRIT/HRIT) global specifications, and the geographical coordinates are calculated based on the World Geodetic System 1984 (WGS84) reference ellipsoid. To convert the disk data to the local geographic coordinate

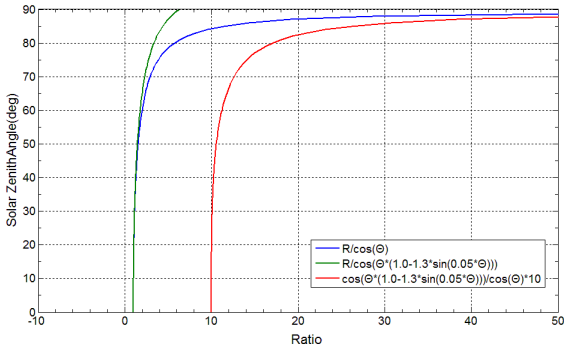


Fig. 2. Comparison of different reflectivity correction methods.

data, projection conversion processing is required. The specific method of converting FY-4B longitude and latitude to row and column number is given as follows.

Converting geographical latitude and longitude of FY-4B to geocentric latitude and longitude of the spherical coordinate system using the following equations [28]:

$$\lambda_e = \mathrm{Lon} \tag{3}$$

$$\varphi_e = \arctan\left(\mathrm{eb}^2/\mathrm{ea}^2 \times \tan(\mathrm{Lat})\right). \tag{4}$$

Calculating the actual distance (re) from the projection point of the disk to the center of the sphere using the following equations:

$$r_e = \frac{eb}{\sqrt{1 - \frac{\mathrm{ea}^2 - eb^2}{\mathrm{ea}^2} \times \cos^2(\varphi_e)}} + H\lambda_e = \mathrm{Lon}. \tag{5}$$

Calculating row and column number

$$c = \mathrm{COFF} + x \times 2^{-16} \times \mathrm{CFAC} \tag{6}$$

$$l = \mathrm{LOFF} + y \times 2^{-16} \times \mathrm{LFAC}. \tag{7}$$

Here, $x = \arctan(-(r_2/r_1))$, $y = \arctan(-(r_3/r_n))$, $r_1 = h - r_e \times \cos(\varphi_e) \times \cos(\lambda_e - \lambda_D)$, $r_2 = -r_e \times \cos(\varphi_e) \times \sin(\lambda_e - \lambda_D)$, $r_3 = r_e \times \sin(\varphi_e)$, and $r_n = (r_1^2 + r_2^2 + r_3^2)^{1/2}$. In the above equations, Lon is the geographic longitude, Lat is the geographic latitude, l is the row number on the nominal map, c is the column number on the nominal map, ea is the long half axis of the Earth, eb is the short half axis of the Earth, h is the distance from the Earth center to the centroid of the satellite, H is the pixel elevation value, λ_D is the longitude of the subsatellite point, COFF is the column offset, CFAC is the column scale factor, LOFF is the row offset, and LFAC is the row scale factor.

c) Fine offset correction: Based on the FY-4B satellite data and the corresponding digital elevation model (DEM) data from January to June 2022, the vector data with obvious geometric characteristics were used for manual analysis and processing. The DEM data were classified by the 500-m interval and were divided into eight intervals for the data with 0–4000 m. Three to five sampling points for each typical coastline and lake without cloud cover in each elevation interval were selected, and finally, a total of 267 points were obtained. The coastline and lake vectors were used as the reference to determine the offset of FY-4B positioning data. According to the divided eight intervals, the longitude offset and latitude

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offset of sampling points in the intervals were counted [29]. The calculation formulas are based on the following equations:

$$\mathrm{Lon}_{\mathrm{delt}} = \mathrm{Lon}_{\mathrm{FY4B}} - \mathrm{Lon}_{\mathrm{shp}} \tag{8}$$

$$\mathrm{Lat}_{\mathrm{delt}} = \mathrm{Lat}_{\mathrm{FY4B}} - \mathrm{Lat}_{\mathrm{shp}} \tag{9}$$

where $\mathrm{Lon}_{\mathrm{delt}}$ and $\mathrm{Lat}_{\mathrm{delt}}$ are the offset of longitude and latitude, respectively; $\mathrm{Lon}_{\mathrm{FY4B}}$ and $\mathrm{Lat}_{\mathrm{FY4B}}$ are the longitude and latitude of FY-4B data, respectively; and $\mathrm{Lon}_{\mathrm{shp}}$ and $\mathrm{Lat}_{\mathrm{shp}}$ are, respectively, the longitude and latitude of the vectors, such as coastline and lake.

2) Principle of Fire Point Identification: The basic principle of remote sensing fire identification is that with the increase of temperature, the thermal radiation increases, and with the increase of thermal radiation, the thermal radiation energy at different infrared channels increases differently. Different objects in nature have different spectral characteristics due to their own physical and chemical properties. When biomass is burned, the main radiation sources are flame, carbons with higher temperature, water vapor, smoke, and so on. According to the Stephen–Boltzmann law [30], the temperature change of high-temperature heat source is very favorable for identifying high-temperature heat source. Based on Wien's displacement law [19], the peak wavelength of surface radiation at normal temperature is about 11 μm. When biomass is burned, the temperature can reach more than 750 K, and the peak wavelength of radiation is about 4 μm. Taking the radiation temperature when the object is not burning as the background radiation temperature, the difference between the combustion radiation temperature and the background radiation temperature can be used to detect the fire in time from the satellite remote sensing information. For the FY-4B/AGRI, the range of mid-infrared (MIR) wavelength is 3.5–4.5 μm, and the range of far-infrared (FIR) wavelength is 10.3–11.3 μm and 11.5–12.5 μm

$$\mathrm{T} \times \lambda\mathrm{max} = 2897.8(\mathrm{K}, \mu\mathrm{m}). \tag{10}$$

The blackbody temperature (T) and the radiation peak wavelength (λmax) are inversely proportional based on (10). The peak wavelength of surface radiation at normal temperature (about 300 K) is around the wavelength range of FIR channel (FY-4B/AGRI channel 13). The combustion temperature of forest fire is generally above 600 K, and the peak wavelength of thermal radiation is close to the MIR channel (FY-4B/AGRI channel 7). Based on the thermal infrared physical characteristics, we can extract fire pixels with thermal anomalies (including simmering fire and naked fire). The temperature range of naked fire is 800–1200 K [31], [32], and the radiation peak wavelength is close to the short-wave infrared channel (FY-4B/AGRI channel 6 and channel 5).

FY-4B infrared channel data are 10 bits, and the quantization level is 0–1023. The spatial resolution of the subsatellite point is 2 km, and the emissivity of the pixel observed by the satellite is the weighted average of the accumulated emissivity of all parts of the ground objects within the pixel range, based on the following equation:

$$N_t = \left(\sum_{i=1}^{n} S_i * N_{Ti}\right) \Big/ S \tag{11}$$

where N_t is the emissivity of the pixel observed by the satellite, T is the brightness temperature corresponding to the emissivity N_t, Si is the area of the ith subregion in the pixel, N_{Ti} is the emissivity of the subregion, T_i is the temperature of the subregion, and $S$is the total area of the pixel.

When the fire point appears in the observation pixel, because of the rapid increase of the emissivity at the fire point, even if the area of the fire point is far less than the resolution of the pixel, it will still cause a rapid increase in the brightness temperature of the mixed pixel in the MIR band. The brightness temperature of the mixed pixel in the FIR channel also increases, but it is much lower than that in the mid-infrared band. Thus, the brightness temperature of the mixed pixel in the mid-infrared channel will be significantly higher than that of the background pixel. However, the difference between the brightness temperature of the mixed pixel and the background pixel in the FIR channel is generally far less obvious than that in the MIR channel. The high temperature in the small subregion i of the pixel where the fire point is located will cause a much higher emissivity increment in the MIR channel than in the FIR channel, making the weighted average increment of emissivity and brightness temperature increment of each channel in the pixel different, from which the difference can be analyzed to extract the fire point information.

3) FY-4B/AGRI Fire Identification Method:

a) Cloud recognition: Cloud coverage is complex and variable and has the greatest impact on the fire identification algorithm [31]. On the one hand, the presence of clouds can easily block the ground fire information, resulting in the attenuation of fire energy entering the sensor and thus leading to misjudgment. On the other hand, the specular reflection generated by clouds may lead to the misjudgment of fire points. Therefore, cloud recognition is an important factor in fire point identification. Based on the MODIS cloud recognition method, the cloud pixel information was extracted according to different optical radiation characteristics of cloud, water, and land pixels. The identification method is shown in Table III [33]. The cloud pixel is recognized when any of them is satisfied.

TABLE III
LIST OF CLOUD PIXEL DETERMINATION

No.	Identification conditions
1	T_7−T_{13}<4 K
2	T_7−T_{13}>20 K & (T_7 < 275 K | T_{13} < 270 K)
3	R_{vis} > 0.28 & θ_{sz} < 70°
4	T_{14} < 265 K
5	T_{13} < 270 K & (T_{13}−T_{14} < 4 K | T_{13}−T_{14} >60 K)

In Table III, T_7, T_{13}, and T_{14} are the brightness temperature values of the MIR channel, the FIR channel, and the FIR split window channels, respectively. R_{vis} is the reflectance value of the VIS channel and θ_{sz} is the solar zenith angle.

b) Judgment on background suspected fire pixel: The pixels located in the first 20% of the high-temperature pixels in the monitoring area are selected as fire point recognition pixels. The pixels in the high-temperature section are identified one by one. The method is based on the following equation:

$$T_7 \geq T_{13} + 100 \times R_{\mathrm{vis}} + 20\ \mathrm{K}. \tag{12}$$

c) Calculation of background temperature: The average temperature of the neighboring pixels of the detected pixels

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is calculated, including the average bright temperature of the MIR channel, the average bright temperature of the FIR channel, the average difference between the MIR and the FIR bright temperature, and the average reflectance of the VIS channel. These values are used as the judgment basis for the confirmation of the fire pixel.

The monitorable pixel in the 7 × 7 pixels surrounding the detection pixel was selected, while the pixels contact to the detection pixel were excluded if the number of monitorable pixels is less than 20% of the total number of pixels in the neighborhood, expand to 9 × 9, 11 × 11, and up to 19 × 19. If the conditions are still not satisfied, the monitoring of this pixel will be abandoned.

The average and standard deviation of MIR brightness temperature, FIR brightness temperature, the difference between MIR and FIR brightness temperature, and visible channel average and standard deviation are calculated for the monitorable pixels in the neighborhood. The calculation formulas are based on the following equations:

$$\delta T_{c_\mathrm{bg}} = \sqrt{\sum_{i=1}^{n} \left(T_{\mathrm{ci}} - T_{c_\mathrm{bg}}\right)^2 / n} \tag{13}$$

$$T_{c_\mathrm{bg}} = \left(\sum_{i=1}^{n} T_{\mathrm{ci}}\right) \Big/ n \tag{14}$$

where c is the serial number of the monitorable pixel; n is the total number of monitorable pixels in the neighborhood; T_{ci} is the mid-infrared brightness temperature, the FIR brightness temperature, the difference between the MIR and FIR brightness temperature, and the visible reflectance value of the identified pixel; T_{c_bg} is the average brightness temperature of surrounding background pixels in MIR and FIR, and the average difference between MIR and FIR brightness temperature; and δT_{c_bg} is the brightness temperature standard deviation of MIR, FIR, and the difference between MIR and FIR. When the standard deviation is less than 2 K, it is set to 2 K. If the standard deviation is greater than 3.5 K, it is set to 3.5 K.

d) Hot spot extraction: The extraction of satellite hot spots mainly depends on the characteristics of MIR thermal anomalies, which can be divided into two categories: absolute conditions and relative conditions. When the pixel satisfies condition (15) or (16), it can be confirmed as a fire pixel. Because of the full solar angle observation of FY-4B geostationary satellite, when calculating the brightness temperature of the background pixels, the solar zenith angle, cloud, and nonvegetation types have a large influence on the calculation results. When the solar zenith angle is greater than 60°, the reflectance correction is easy to be distorted. To reduce misjudgment, it is necessary to increase the judgment threshold

$$T_7 > 360\ \mathrm{K}\ \&\ R_{\mathrm{vis}} < 0.7 \& \theta_{\mathrm{sz}} > 3^{\circ}. \tag{15}$$

$$T_7 \geq T_{7_\mathrm{bg}} + a \times \delta T_{7_\mathrm{bg}} \& T_{7_13} \geq T_{7_{13_{\mathrm{bg}}}} + a \times \delta T_{7_13_\mathrm{bg}} \tag{16}$$

$$a(P_v, P_c, \theta_{\mathrm{sz}}) = \begin{cases} (\sin\theta_{\mathrm{sz}} + 1) \times (1 + P_v) \times (1 + P_c) \\ \theta_{\mathrm{sz}} < 60^{\circ} \\ (1.2\sin\theta_{\mathrm{sz}} + 1) \times (1 + P_v) \times (1 + P_c)^2 \\ \theta_{\mathrm{sz}} \geq 60^{\circ} \end{cases} \tag{17}$$

where a is a coefficient function, which is related to the proportion of nonvegetation pixels, cloud pixels, and solar zenith angle; P_v is the proportion of nonvegetation pixels in the region; and P_c is the proportion of cloud pixels in the region.

e) Hot spot reprocessing: The thermal anomaly signals of the MIR channel obtained by satellite monitoring include real fire points, specular reflection pseudo thermal signals of cloud edges and high reflection underlying surfaces, and nonfire thermal anomaly points such as bare ground, artificial buildings, and factories. Therefore, it is necessary to do reprocessing on the extracted heat spots and eliminate the nonfire heat anomalies.

If the fire pixel satisfies the following conditions, it is considered to be contaminated by clouds and will not be considered as a fire pixel. The calculation formulas are based on the following equation:

$$R_{\mathrm{vis}} \geq R_{\mathrm{vis}_{\mathrm{bg}}} + RC_\mathrm{th}\ \&\ \ T_{13} \leq T_{13_\mathrm{bg}} - \mathrm{TC_th}. \tag{18}$$

Here, RC_th is the recognition threshold of VIS cloud pollution, and the initial value is set to 0.15. TC_th is the threshold of FIR cloud interference with the initial value of 5 K.

If the pixel satisfies the following conditions, it is considered as the cloud edge or the desert edge, which will not be regarded as the fire pixel. The calculation formulas are based on the following equation:

$$T_7 \leq T_{7_\mathrm{bg}} + \mathrm{C} \times \delta T_{7_\mathrm{bg}}\ \&\ T_{7_13_\mathrm{bg}} \leq T_{7_13_\mathrm{bg}} + \mathrm{C} \times \delta T_{7_13_\mathrm{bg}} \tag{19}$$

where C is the threshold of cloud edge and desert edge identification, and the initial value is set to 8 K.

f) Supplementary judgment of fire in cloud area: Fire identification in cloud area is mainly used to identify the fire under thin clouds.

When the cloud pixel satisfies the following condition equation, the average brightness temperature of the cloud area is calculated:

$$T_7 \geq \mathrm{TC}_7\mathrm{th}\ \&\ T_{7_13} \geq T_{7_13}\mathrm{th}\ \&\ R_{\mathrm{vis}} \leq RC_{\mathrm{vis}}\mathrm{th} \tag{20}$$

where TC_7th is the infrared brightness temperature threshold of cloud pixel, with the initial value being 325 K; T_{7_13}th is the threshold of brightness temperature difference of cloud pixel between MIR and FIR channels, with the initial value set to 30 K; and RC_{vis}th is the VIS reflectance threshold of cloud pixel, with the initial value set to 0.6.

The fire pixels in cloud region were identified using the following equation:

$$T_7 \geq T_{7_\mathrm{bg}} + T_{\mathrm{fire}}\mathrm{th}. \tag{21}$$

Here, $\mathrm{TC}_{\mathrm{fire}}$th is the threshold of fire pixel identification in the cloud area, with the initial value set to 20 K.

4) Estimation Method of Naked-Fire Area: The subpixel naked-fire area is estimated by using the single-channel subpixel fire area estimation method [20] (22). When the brightness temperature of the MIR channel is less than 750 K, the MIR channel estimation is used; when the brightness temperature of the MIR channel is greater than or equal

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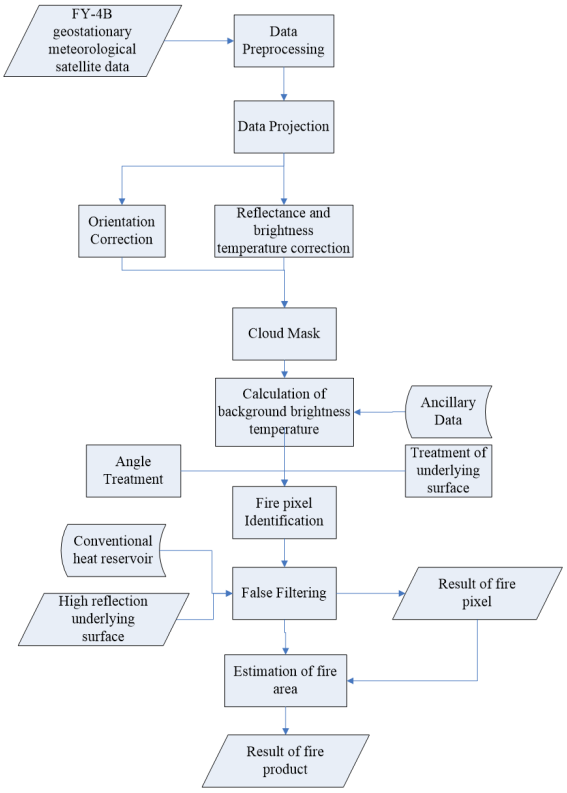


Fig. 3. Flowchart of the fire identification by the FY-4B/AGRI.

to 750 K, the FIR channel estimation is used in the following equation:

$$S_f = P \times S \tag{22}$$

$$P = (L_{\mathrm{MTR_mix}}(T_{\mathrm{MTR_mix}}) - L_{\mathrm{MTR_bg}}(T_{\mathrm{MTR_bg}})/(L_{\mathrm{MTR}}(T_{\mathrm{f}}) - L_{\mathrm{MTR_bg}}(T_{\mathrm{MTR_bg}}) \tag{23}$$

$$P = (L_{\mathrm{FTR_mix}}(T_{\mathrm{FTR_mix}}) - L_{\mathrm{FTR_bg}}(T_{\mathrm{FTR_bg}})/(L_{\mathrm{FTR}}(T_f) - L_{\mathrm{FTR_bg}}(T_{\mathrm{FTR_bg}}) \tag{24}$$

where S_f is the subpixel fire spot area, P is the subpixel fire spot area ratio, and S is the pixel area. T_f is the subpixel fire point temperature, set to 750 K. $L_{\mathrm{MIR_mix}}$ and $L_{\mathrm{FAR_mix}}$ are the mixed pixel emissivity of the MIR and the FIR channels, respectively. $L_{\mathrm{MIR_bg}}$ and $L_{\mathrm{FAR_bg}}$ are the background pixel emissivity of MIR and far-infrared channels, respectively. $T_{\mathrm{MIR_mix}}$ and $T_{\mathrm{FAR_mix}}$ are the brightness temperature of the mid- and far-infrared mixed pixels, respectively. $T_{\mathrm{mid_bg}}$ and $T_{\mathrm{far_bg}}$ are the brightness temperature of the background pixel in the mid- and far-infrared channels, respectively.

5) FY-4B/AGRI Fire Monitoring Procedure: To realize the automatic fire point identification of FY-4B, the fire point monitoring algorithm used by FY-4B is improved based on the characteristics of FY-4B sensor channel [16]. In addition, the algorithm also considers the influence of abnormal high-temperature pixels on fire pixel extraction, such as conventional heat source points and cloud edge heat source points, and eliminates abnormal heat source points (nonfire pixels). Fig. 3 shows the flowchart of fire point identification procedure.

TABLE IV
CENTRAL WAVELENGTHS OF SOME CHANNELS OF FY-4B AND GK-2A

	FY-4B/AGRI Channel (Center Wavelength,um)	GK2A/AMI Channel (Center Wavelength)
VIR004	#1(0.47)	#1(0.47)
NR008	#3(0.83)	#2(0.86)
IR038_H	#7(3.75)	#5(3.80)
IR038_L	#8(3.75)	#5(3.80)
IR108	#13(10.8)	#6(10.5)
IR120	#14(12.0)	#7(12.3)

VIR： Visible light； NR：Near infrared；IR：Infrared

III. RESULTS AND DISCUSSION

A. FY-4B/AGRI Data Quality Inspection

In order to test the data quality of FY-4B/AGRI, this study cross-compares the channel data related to fire point identification of FY-4B/AGRI and Advanced Meteorological Imager on GK-2A (GK-2A/AMI). Table IV shows the comparison of the central wavelengths of some channels between FY-4B/AGRI and GK-2A/AMI. GK-2A is a new generation of geostationary meteorological satellite in South Korea, which was successfully launched on December 4, 2018 and is located over the equator at 128.2°E. Its product and data quality have been widely recognized internationally [16], [17], [18]. The positioning and channel of FY-4B are similar to GK-2A. Thus, the data quality of FY-4B can be checked by cross comparison with GK-2A channels. Fig. 4 shows the density scatter plots of satellite channel reflectance and brightness temperature parameters. As can be seen, the parameters of AGRI and Advanced Meteorological Imager (AMI) channel are highly consistent. The root-mean-square error (RMSE) of reflection channel is only 1.33%, and the maximum RMSE of brightness temperature channel is less than 1.3 K. It indicates that the performance of this channel can better satisfy the application of forest and grassland fire monitoring.

The resolution of FY-4B/AGRI MIR channel is 2 km. For the identification and fine positioning of small fire spots, subsequent statistical analysis of large samples is required. FY-4B has a small amount of historical data, and its sensitivity of fire point identification and the accuracy of fire point positioning need further research.

B. Analysis of Longitudinal and Latitudinal Offsets of FY-4B/AGRI Data in Different Elevation Intervals

Through the positioning analysis of different elevation intervals, we find that the average longitude offset of FY-4B satellite data is −0.5° to 0° and the average latitude offset is −0.9° to 0.6°. Table V shows the offset statistics of different intervals. It indicates that the longitude and latitude of FY-4B data are smaller than the real positioning. The higher the altitude is, the larger the offset is. The offset is less than 0.01° for the DEM less than 1500 m and more than 0.02° for the DEM over 3000 m.

C. Application of FY-4B/AGRI Data in Grassland Fire Monitoring

On April 18–20, 2022 BJT, a grassland fire occurred in Mongolia adjacent to the border of Xilingol League in Inner

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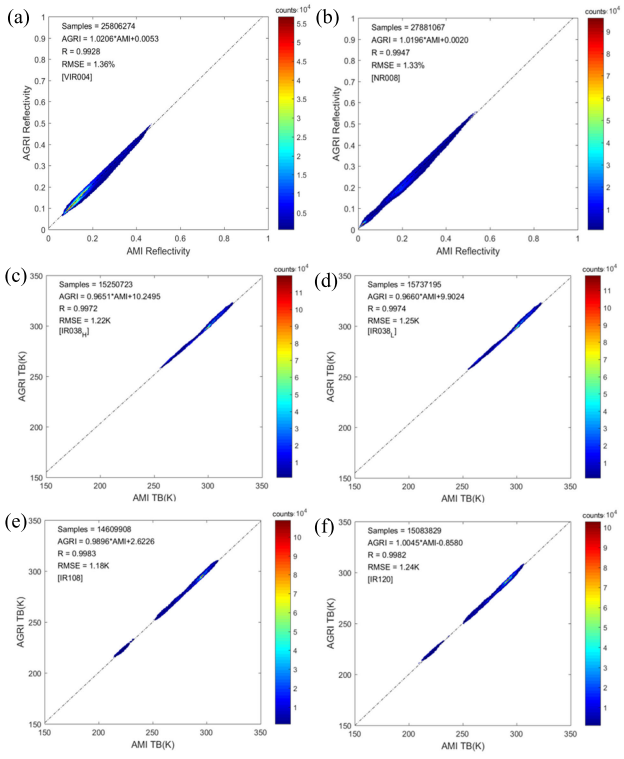


Fig. 4. Comparison of data between FY-4B/AGRI and GK-2A/AMI: (a) VIR004, (b) NR008, (c) IR038_H, (d) IR038_L, (e) IR108, and (f) IR120.

TABLE V
LONGITUDE AND LATITUDE OFFSETS IN DIFFERENT ELEVATION INTERVALS

DEM(m)	Lon_{delt}(°×100)	Lat_{delt}(°×100)
0~500	-0.7	-0.5
500~1000	-0.9	-0.6
1000~1500	-1.0	-0.7
1500~2000	-1.2	-0.9
2000~2500	-1.5	-1.1
2500~3000	-2.2	-1.8
3000~3500	-2.5	-2.0
3500~4000	-2.7	-2.1

Mongolia Autonomous Region of China. The burned area exceeded 2500 km^2, and it was the most threatening grassland fire event in the border of Inner Mongolia Autonomous Region of China in recent years. The FY-4B geostationary satellite fire point identification algorithm proposed in this study was used to continuously track and monitor the panoramic and entire process of this grassland fire event. The dynamic development information of this grassland fire is obtained, including the area of naked fire at each observation time, the temporal and spatial dynamic change range of the fire site, and the spatial distribution of high-intensity fire areas in the fire site.

1) Spatiotemporal Dynamic Analysis of Fire Spread: In this study, fire point information was extracted from FY-4B/AGRI for a total of 35 h from 15:00 BJT on April 18 to 00:00 BJT on April 20, 2022. Fig. 5(a) shows the time series of fire point observation. As shown in the figure, the fire started at 15:00 BJT on April 18 near the border of Xilingol League in Inner Mongolia Autonomous Region of China (dark blue in the figure) and then spread eastward. At around 15:00 BJT on

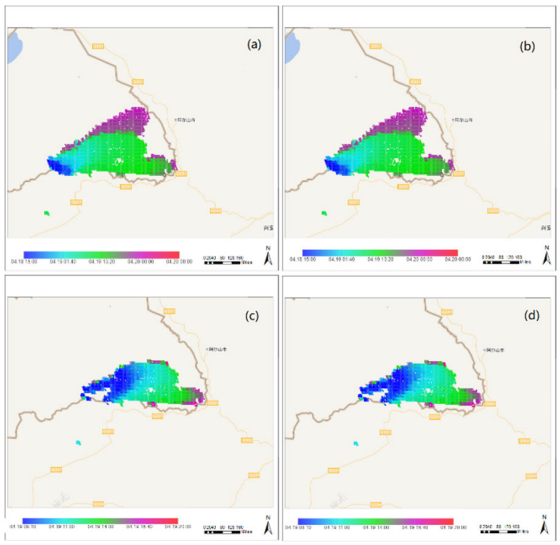


Fig. 5. Spatiotemporal distribution of fire site in April 2022: (a) whole time period, (b) from 15:00 BJT on the 18th to 06:00 BJT on the 19th, (c) from 06:00 BJT to 17:00 BJT on the 19th, and (d) from 17:00 BJT on the 19th to 00:00 BJT on the 20th.

April 19, the fire spread to near the border of Xing'an League. The east-to-west distance of the fire site was about 93 km, and the largest north-to-south distance was about 61 km, forming a wide-range fire area.

The spreading process monitored by the FY-4B satellite shows that the grassland fire can be divided into three stages. The first stage [Fig. 5(b)] is from 15:00 BJT on April 18 to 06:00 BJT on April 19, and this is the fire initial stage. In this stage, the fire spread eastward and northward with the wind. The second stage [Fig. 5(c)] is the fire rapidly spreading stage, and it is from 06:00 BJT to 17:00 BJT on April 19. In this stage, the direction of fire spread changed from northeasterly to southeasterly, and it quickly spread to the border of Xi'an League and Xing'an League in China, forming a wide-range fire area. The third stage [Fig. 5(d)] is the fire spreading direction changing stage and is from 20:00 BJT on April 19 to 00:00 BJT on April 24. In this stage, the northern part of the fire spread northward and eastward, reaching the border of Xing'an League and Hulunbuir League. At the same time, there was still a small amount of naked fire in the southeast of the fire area.

Because of the homogeneous underlying surface of grassland, the grassland fire generally spreads over a large area, which is conducive to monitoring the spatiotemporal distribution of fire spreading using geostationary satellite data. However, forest fires occur more frequently in China. Compared with grassland fire, its underlying surface varies greatly and its spreading area is relatively small. How to apply FY-4B satellite data to forest fire prevention will be the focus of our future work.

2) Dynamic Variation Analysis of Naked-Fire Area: According to the continuous monitoring results of FY-4B, the naked-fire area was extracted hour by hour. The variation of naked-fire area shows that there are two peaks of naked-fire area in this fire episode, namely, 09:00 BJT and 21:00 BJT on the 19th (Fig. 6). In the first stage of the fire episode, the naked-fire area gradually increased, but the growth rate was relatively small. The maximum naked-fire area appeared at

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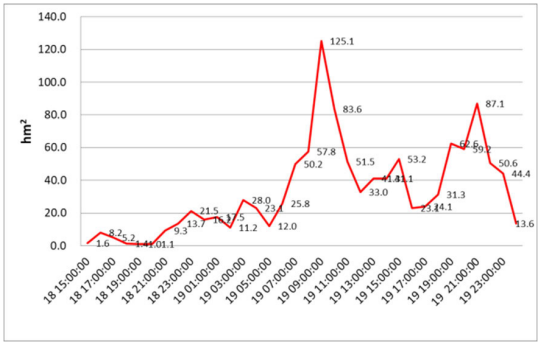


Fig. 6. Estimated area of naked fire from 15:00 BJT on April 18 to 00:00 BJT on April 20, 2022.

04:00 BJT on the 19th, about 28 km^2. In the second stage, the naked-fire area increased rapidly. The maximum value appeared at 09:00 BJT on the 19th, about 125 km^2, and then gradually decreased. In the third stage, the fire area increased again, and the second peak (about 87.1 km^2) appeared at 21:00 BJT on the 19th. From the analysis of fire development, in the first stage, because the fire has just occurred and the fire was in the gestation period, the basic overfire area was small, and the increase in the overfire area caused by the fire spreading is small. As seen in Fig. 4(c), the second stage is the fire rapidly spreading stage, with a spreading area exceeding 180 000 km^2. The maximum instantaneous naked-fire area occurred in this period. In the third stage, except for the fire spreading in the north, there were only some sporadic fire spots in the original fire area, and the overall fire area was significantly smaller than that in the second stage.

3) Analysis of High-Intensity Fire Spot Distribution: To further analyze the fire spot monitoring quality of FY-4B/AGRI and to obtain information about the fire spots with different intensities (especially high-intensity fire spots), this study selects the typical time period to represent the start, development, and other different stages of the fire. The original satellite observation images covering the panorama of the fire area at 15:00 BJT, 19:00 BJT, and 23:00 BJT on April 18th and at 06:00 BJT, 09:00, 13:00 BJT, 17:00 BJT, and 21:00 BJT on April 19th, 2022 were extracted.

For the infrared channel signal of fire monitoring, the temperature and the central wavelength are inversely proportional to each other according to the Wien displacement law, and the energy of different channels changes with the change of fire. In this study, the combination of MIR, FIR, and short-wave infrared was used, that is, the three-channel composite images of 3.7, 11, and 2.23 μm were used to maintain the consistency of daytime observation and to reveal the distribution of different fire intensities in the fire field.

The fire monitoring diagram is shown in Fig. 7. As can be seen, the fire scene display colors are mainly pink, yellow, and white. Pink represents the single MIR channel signal, yellow represents the MIR and FIR channel signal, and white represents the comprehensive signal of the three channels. When the fire is small, the thermal anomaly is mainly reflected in the MIR channel [Fig. 7(a)]. For example, in the initial stage of this fire episode, the fire was not strong, the energy was mainly in the 3.7-μm channel, and the fire point shows pink color. As the fire intensifies, both FIR and near-infrared NIR channels showed thermal anomalies [Fig. 7(c)]. Thus, the periphery of fire spot is in pink and the center is in yellow.

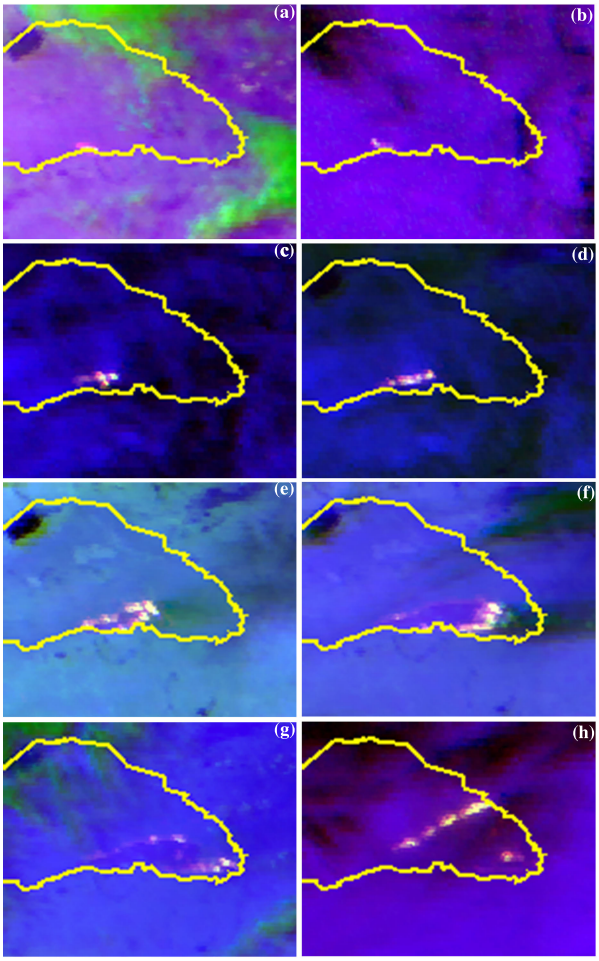


Fig. 7. FY-4B fire spot monitoring map: (a) 15:00 April 18, (b) 19:00 April 18, (c) 23:00 April 18, (d) 06:00 April 19, (e) 09:00 April 19, (f) 13:00 April 19, (g) 17:00 April 19, and (h) 21:00 April 19.

When the fire intensifies to a certain extent, the 2.22-μm short-wave infrared will show a thermal signal, and the fire spot will be in white. As shown in Fig. 7(f), the central area of the fire tongue (fire spreading head) is in white, the surrounding pixels are in yellow, and the periphery of the fire site shows a pink contour, that is, the strongest fire intensity is at the tongue of fire and the weaker fire intensity is at the fire edge. These finds show that the sensitivity characteristics of FY-4B/AGRI different channels to different fire point intensities can be effectively applied to the disaster prevention and emergency support services of forest and grassland fire.

Different channel combinations show different fire intensity distributions. This information not only contains fire intensity but also contains refined positioning of fire points, which is of great significance for fire prevention. Therefore, through different infrared channel combinations, fire information extraction can be further quantitatively studied. Further research can be conducted to extract fire information by using FY-4B/AGRI multiband infrared channel data.

IV. Conclusion

In this study, based on the FY-4B/AGRI data, the performance of the instrument, data quality, fire extraction algorithm, and its application in dynamic monitoring of grassland fire were investigated separately. The main conclusions are given as follows.

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The instrument parameters of FY-4B/AGRI have reached the advanced level of similar meteorological satellites in the world. Through cross comparison with GK-2A/AMI data, we find that the data quality of FY-4B/AGRI is basically the same as that of the internationally leading GK-2A.

The fire identification algorithm of FY-4B is based on the observation characteristics of geostationary satellites. Based on more mature fire identification algorithms such as MODIS, Fengyun-3-D Medium Resolution Spectral Imager (FY-3-D/MERSI), and Himawari-8Advanced Himawari Imager (Himawari-8/AHI), it considers the influence of all solar angle observation and underlying surface information such as cloud and vegetation coverage. In addition, the reprocessing steps of geographic positioning, reflectance fine correction, and eliminating false fire points are added, which further improves the accuracy, stability, and practicability of the algorithm.

Through the case outside Mongolia from April 18 to 20, 2022, the fire monitoring advantages of FY-4B, such as high observation frequency and multiband data, are demonstrated by using the minute observation characteristics of FY-4B. The dynamic continuous monitoring of the fire situation and spatial spreading change in the fire field is realized.

The setting and combination of FY-4B/AGRI channels can not only obtain fire location information but also finely extract different fire distributions in the fire site, providing quantitative fire-fighting decision information for front-line fire prevention personnel.

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